

WHAT IT DOES

Keeps the tent at the target climate for the selected grow profile. Every 30 s it reads temperature, humidity and CO2, scores how far each is off ideal (0-100, 50 = on target), and commands all actuators at once from that score.

Sensor and event data is written to the SD card; if the card is missing it falls back to internal flash, then to RAM, and migrates the backlog once the card returns.

BUTTON

Tap	next screen
Hold 3 s	screen action
Idle 30 s	back to screen 1
Idle 120 s	display sleeps
Any press	wakes display

PHASENWECHSEL (grow phase)

The grow phase advances on its own after so many weeks. The screen then shows PHASENWECHSEL with the new humidity target, the new light level, and whether the humidifier still runs.

It stays up until you press the button - through sleep and power cuts - and the Service LED stays on until you do. The first press on a dark screen only wakes it; press again to confirm.

OLED SCREENS (tap to cycle)

TEMPERATURE	now/hi/lo/avg, 1 h. Hold = clear
HUMIDITY	now/hi/lo/avg, 1 h. Hold = clear
SERVICE	days since service. Hold = reset
SD CARD	mount state + MB. Hold = remount
ALERTS	active ERR/WRN keys, or All OK
SYSTEM	clock, firmware, uptime, RAM
RELAYS	mains channels + PWM fan duty
REGULATION	title = grow phase; band, commands
CO2	ppm + vent override
SOIL	moisture % + root temperature
DEBUG	Hold = open test menu

REGULATION SCREEN

Title REG bloom - the grow phase now running (seedling / stretch / bloom). Plain REGULATION means no phase plan.

Line 1 Band 2 ok - severity band, and mode: ok / EMERG (emergency vector) / LATCHED (held in safe state, needs a power cycle or a fixed climate to clear).

T H C	deviation temp / humidity / CO2
	0 = far low, 50 = ideal, 100 = far high

Lines 3-5 are commanded output, 0-100 %:

He	heater
Fo	heater follower fan
Cl	cooler
Hu	humidifier
Ex	exhaust fan
Ci	circulation fans
Gl	grow light
b	day/night blend, 0 = night, 1 = day

MAINS RELAYS (REL_CON1)

Cool GP18	cooler / AC
Humi GP19	humidifier
Lite GP20	grow light (+ 0-10 V dimmer)
Spar GP21	spare, unassigned
Screen shows ON / OFF, or -- when the channel has no controller.	

PWM FANS (PCA9685, 12 V)

ch0	circulation, centre
ch1	circulation, walls
ch2	heater follower
ch3	case fan, always on 60 %
ch4	exhaust

STATUS LEDS

Activity	blinks on each read/write
SD	off = mounted, solid = no card, blink = card present but unreadable
Warning	solid = degraded, still running
Error	solid = fault, needs attention
Service	blinks when service is due
Pico LED	heartbeat, firmware is alive

SENSORS

SHT31-D	temperature + humidity, I2C
Senseair S8	CO2 ppm, UART
DS3231	real-time clock, I2C
Soil STEMMA	moisture + root temp, I2C

ALERT KEYS (ALERTS screen)

ERR sd_required	SD needed but unavailable
ERR th_dead	climate sensor not answering
ERR mem_error	RAM critically low
ERR logged_error	see system.log on the SD card
WRN th_intermit	sensor dropping reads
WRN fallback_act	logging to flash, not SD
WRN buffer_backlog	writes queueing up
WRN mem_warn	RAM above warn level
WRN rtc_invalid	clock time implausible
WRN co2_stale	CO2 reading too old; timed venting
WRN co2_unreach	CO2 sensor silent; check its plug
WRN sht31_unreach	climate sensor silent; check its plug
WRN humidifier_in	humidifier runs, air stays dry: refill it
WRN rh_target_unr	room air too humid for the target
WRN soil_unreach	soil probe silent; check its plug
WRN root_temp_low	root zone under 20 C
WRN root_temp_hig	root zone over 26 C

FIRST AID

PHASENWECHSEL	not a fault; press to confirm
SD LED on	reseat the card, then SD screen, hold
LATCHED	fix climate, then power cycle
Blank screen	tap the button; it sleeps at 120 s
Nothing runs	check 12 V input and the fuse